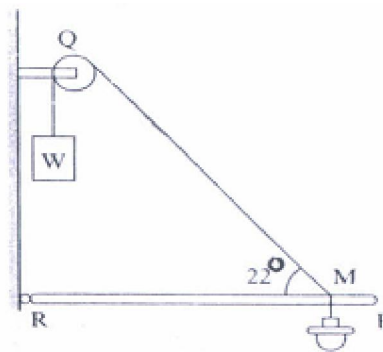


MODEL QUESTIONS FOR PRACTICE

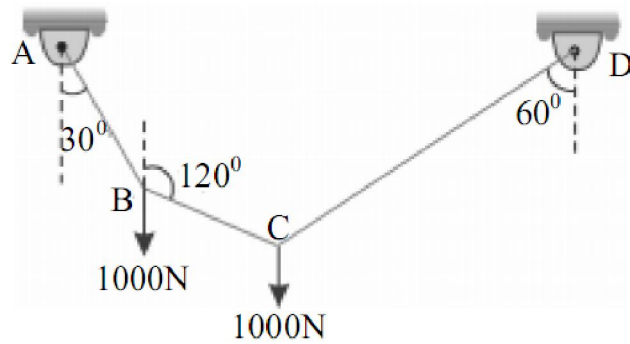
ME 101: BASIC MECHANICAL ENGINEERING

Module 1: System of Forces and Structural Mechanics

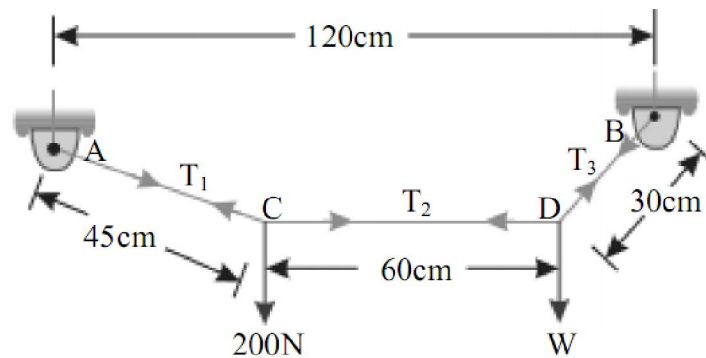
1. How will you obtain the resultant of a coplanar, non-concurrent force system?
2. What is a free-body diagrams?
3. What is principle of transmissibility of a force?
4. Sketch a non-concurrent, non-parallel-coplanar force system.
5. Define moment and moment of force.
6. What is a couple? Enumerate its various characteristics.
7. Define truss and differentiate between perfect, deficient and redundant trusses.
8. Differentiate between truss and frame.
9. Write down steps for finding out the forces in the member of a truss by the method of joints.
10. State the assumptions made while making an analysis of a truss.
11. Define stress and strain.
12. Explain the concept of stress.
13. Draw stress-strain curve for a ductile and brittle material on a simple diagram. What are the differences between these two curves?
14. State Hooke's law.
15. Define Poisson's ratio, shear stress, volumetric strain, thermal stress. yield points and tensile strength.
16. Define E, G and K
17. Establish relationship among E, G and K.
18. State Hooke's law. Distinguish between limit of proportionality and elastic limit.
19. An electric light fixture is held with the arrangement shown in Figure. If the weight of the fixture is 20 kg and the hinge is an Ideal one, determine the axial force in the bar and the weight W suspended from pulley Q required to hold the string stable.



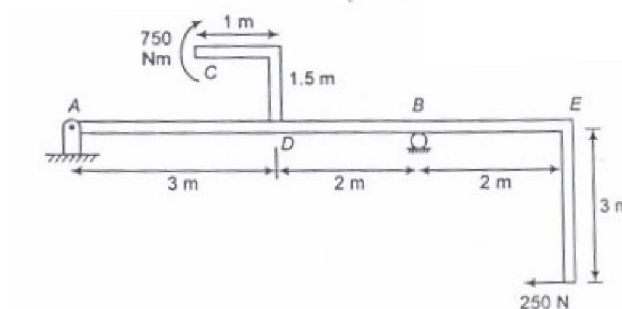
20. A string ABCD, attached to fixed points A and D has two equal weights of 1000 N attached to it at B and C. The weights rest with the portions AB and CD inclined at angles as shown in figure. Find the tensions in the portions AB, BC and CD of the string, if the inclination of the portion BC with the vertical is 120° .



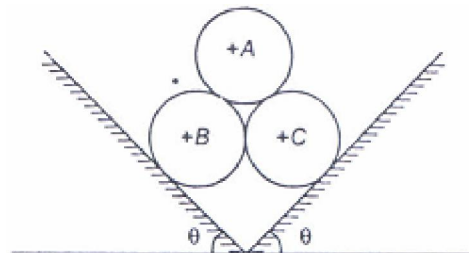
21. A rope is connected between two points A and B, 120 cm apart at the same level. A load of 200 N is suspended from a point C on the rope 45 cm from A as shown in figure. Find the load that should be suspended from the rope D 30 cm from B, which will keep the rope CD horizontal.



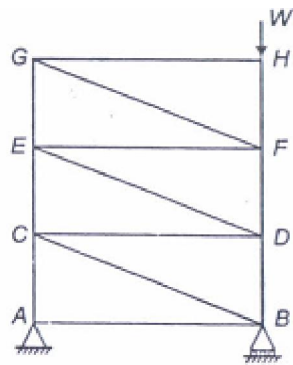
22. Determine the reaction at supports for the bracket shown in Figure.



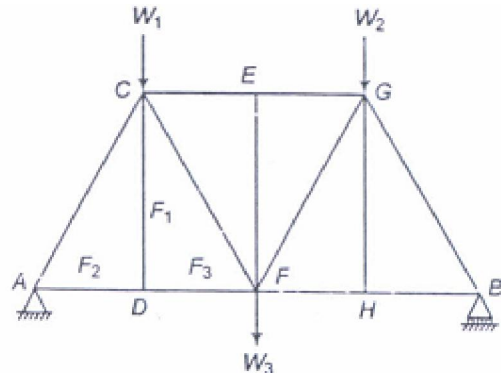
23. Three identical cylinders each weighting W stacked as shown in Figure. On smooth incline surfaces each inclined at an angle θ with horizontal. Determine the smallest angle θ to prevent the stack from collapsing.



24. Identify the members having zero force.



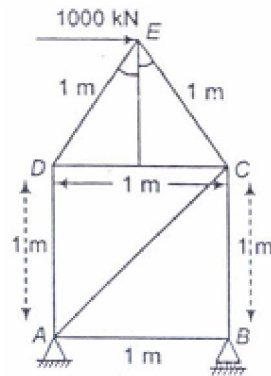
(a)



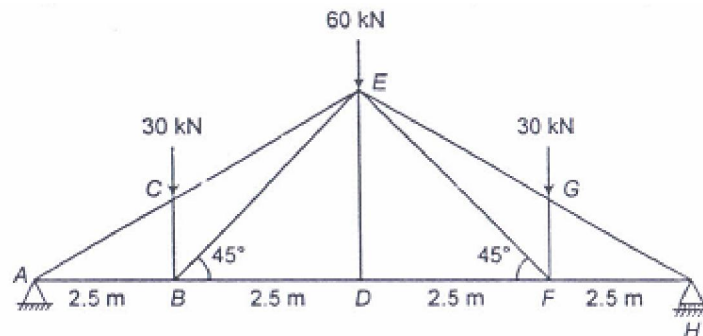
(b)

25. Determine the force in member AC using

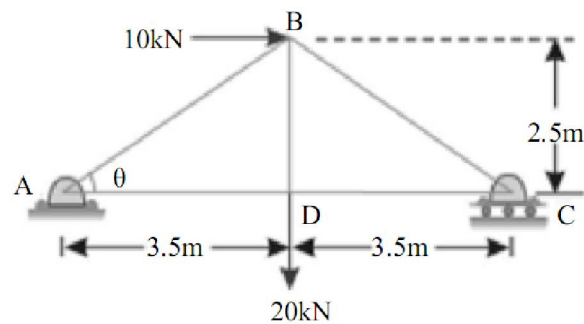
i) The method of Joints, (ii) The method of Sections.



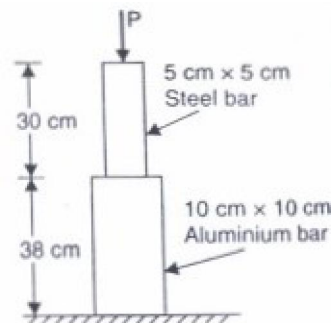
26. Find the forces in members BE, CE and BD for the truss by the method of sections.



27. Figure shows a truss of 7m span and 2.5m height subjected to two-point loads at B and D. Determine the forces in all the members of the structure.



28. A member formed by connecting a steel bar to an aluminium bar is shown in figure. Assuming that the bars are prevented from buckling sideways, calculate the magnitude of force that will cause the total length of the member to decrease 0.25 mm. The values of elastic modulus for steel and aluminium are $2.1 \times 10^5 \text{ N/mm}^2$ and $7 \times 10^4 \text{ N/mm}^2$ respectively.



29. A 30 m length of railway is to be so laid that there is no stress in the rails at 10°C . Determine the stress induced in the rails at 60°C if
- if there is no allowance for expansion
 - if there is an expansion allowance of 8 mm per rail
- (b) What should be the expansion allowance if the stress in the rail is to be zero at 60°C temperature?
- (c) What is the maximum allowable temperature to have no stress in the rails if expansion allowance is 12 mm per rail?
- (Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\alpha = 12 \times 10^{-6} \text{ per deg C}$.)
30. A rod 10m long at 10°C is heated to 70°C . If the free expansion is prevented, find the magnitude and nature of stress induced. Take $E = 2.1 \times 10^5 \text{ N/mm}^2$, $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$.
31. The temp of the composite system is raised by 60°C . Find out the stresses in three portion the bar (a) If the supports are rigid (b) The supports yield by 0.05 cm.
- Given:

